Hyperspectral Image Super-Resolution With ConvLSTM Skip-Connections

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***Abstract*— Hyperspectral image super-resolution has been extensively studied, and significant development has been made based on deep convolutional neural networks (CNNs). Particularly, residual networks that fuse features from multiple layers have achieved high-accuracy hyperspectral image super-resolution. However, most residual networks tend to straightforwardly add features from one layer to another through skip-connections that may cause confusion about feature fusion. To tackle this issue, we develop a ConvLSTM skip-connection strategy that characterizes features from consecutive layers by ConvLSTMs and renders feature fusion in a more principal manner. Accordingly, we develop a super-resolution framework that consists of three modules. The first module, i.e., spatial feature reconstruction, employs the ConvLSTM skip-connections to comprehensively fuse spatial features from different layers. The second module, i.e., edge refinement, involves the ConvLSTM skip-connections to enhance the edge information from intermediate results. The third module, i.e., spectral information reconstruction, refines spectral features by capturing interactions between different spectral bands through the ConvLSTM skip-connections. The three complementary modules cooperate such that both spatial resolution and spectral fidelity are well maintained. Extensive experimental results on the Chikusei, Houston, and QUST-1 datasets demonstrate that our framework outperforms state-of-the-art methods in terms of quantitative evaluation and visual quality across a variety of scenarios. We release our source code at https://gitee.com/xu_yinghao/CLSCNet for public evaluations.**

***Index Terms*— ConvLSTM, edge enhancement, feature fusion, hyperspectral image, super-resolution.**

I. Introduction

HYPERSPECTRAL images are typically composed of dozens to hundreds of spectral bands [1]. Their rich spectral information accurately characterizes the internal structure and material properties of objects on the Earth [2]. Hyperspectral images are widely applied in various scenarios, including geological exploration [3], city analysis [4], and land change detection [5]. Zhang and Zhang [6] provided a comprehensive review of the recent achievements of artificial intelligence algorithms and applications in remote sensing data analysis. In addition, with the development of deep learning, foundation models have been developed for hyperspectral remote sensing [7], [8], [9]. In these tasks, there has been an increasing requirement for both high spatial resolution and rich spectral details. Hyperspectral image super-resolution is an important technique for improving hyperspectral images' spatial resolution. It transforms low spatial resolution hyperspectral images into high spatial resolution ones, without altering physical equipment. It is thus a very useful technique for hyperspectral image applications.

A. Literature Review of Hyperspectral Image Super-Resolution

Hyperspectral image super-resolution methods can be classified into two categories according to different image types, i.e., fusion-based super-resolution [10], [11], [12] (also known as pansharpening) and single-image super-resolution [13], [14], [15]. Fusion-based super-resolution is a technique that reconstructs high spatial resolution hyperspectral images. The method for its reconstruction involves the fusion of high spectral resolution images with their corresponding high spatial resolution images. Zare et al. [16] proposed a novel image fusion approach based on coupled nonnegative tensor decomposition (NTD). The NTD is directly applied to the fusion process of the hyperspectral image and multispectral image, thereby achieving the complete preservation of the spatial and spectral structure of hyperspectral images. Hu et al. [17] proposed a transformer-based architecture specifically designed for hyperspectral image fusion super-resolution. It utilizes the self-attention mechanism to globally explore the intrinsic relationships within fusion features for the purpose of addressing the limitations of local neighborhood relations in CNN-based techniques. Cao et al. [18] proposed an interpretable deep network structure for pansharpening, which adopts a model-driven approach by utilizing convolutional sparse coding and an alternative algorithm to optimize the pansharpening model. Zhang et al. [19] established a deep multisource convolutional sparse coding model. It integrates multisource

Manuscript received 5 April 2024; revised 3 May 2024; accepted 6 May 2024. Date of publication 16 May 2024; date of current version 31 May 2024. This work was supported in part by the Major Program of the National Natural Science Foundation of China under Grant 61890960, in part by the National Natural Science Foundation of China under Grant 62375027, and in part by the Taishan Scholar Project. *(Corresponding author: Peng Ren.)*

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Digital Object Identifier 10.1109/TGRS.2024.3401843

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convolutional sparse coding with degradation models and deep priors to fully capture the shared common information between high spatial resolution multispectral images and panchromatic images. Ciotola et al. [20] proposed a deep learning-based pansharpening model. It utilizes residual attention modules and a loss function suitable for pansharpening, jointly improving the spectral and spatial quality of pansharpened data. These fusion methods establish relationships between two inputs to enhance the quality of super-resolution images. Single-image super-resolution reconstructs image spatial information without the need for additional data. Its inherent flexibility and applicability have garnered significant attention in practical applications.

Existing super-resolution methods for single images are categorized into two groups [21], i.e., traditional methods and deep learning methods. Traditional image super-resolution mainly includes methods based on interpolation, regularization, and sparse representation. These methods utilize different prior knowledge to reconstruct image spatial information from limited data. Wang et al. [22] utilized total variation norm regularization to ensure structural smoothness in super-resolution. Wang et al. [23] proposed a multidirectional dictionary sparse representation for hyperspectral image super-resolution. They classified patches in the depth map based on their geometric directions and trained separate dictionaries to enhance sparsity. Due to the limited representational capacity of their manually crafted features, these methods were insufficient to recover much detail, resulting in low-quality image reconstruction.

In recent years, deep learning has emerged as an effective approach to achieving single-image super-resolution. Dong et al. [24] proposed a super-resolution convolutional neural network (CNN), demonstrating its superiority over traditional methods in the nonlinear mapping from low-resolution information to high-resolution information. Subsequently, residual networks have been widely employed in single-image super-resolution tasks [25], [26], [27], achieving remarkable results [28], [29], [30], [31], [32]. The effectiveness is primarily attributed to the ability of the residual networks to represent multilevel features and fuse them for super-resolution. Guo et al. [33] used a dense nonlocal and local block structure to combine local and global features to reconstruct hyperspectral images with fine details. Dong et al. [34] proposed a context-aware guided attention-based cross-feedback dense network, which employed a feedback mechanism across two branches to provide reliable spatial–spectral guidance for low-level features. Li et al. [35] employed a full-band image input to generate images using a generative adversarial network, resulting in visually more realistic images. Liu et al. [36] proposed a spectral grouping and attention-driven residual dense network, which explores spatial–spectral features through spectral grouping and attention-driven residual dense blocks. While the CNNs with 2-D convolution demonstrate excellent performance in handling spatial information of hyperspectral images, they fall short in learning spectral information. This leads to the disorderliness of spectral information. To address this issue, 3-D convolution for learning spectral information has been explored. Mei et al. [37] proposed a 3-D full CNN (3DFCNN), which uses 3-D convolution to directly extract spatial context information from neighboring pixels. This method extracts the spectral correlation from adjacent bands, thus alleviating the spectral distortion of the generated images. Arun et al. [38] designed a single image super-resolution architecture based on 3-D CNNs, which is capable of adjusting the spatial–spectral features of hyperspectral cubes to finer spatial resolution. It is important to note that in 3-D networks, there are typically a large number of parameters whose training requires considerable computational overheads. To alleviate this issue, Huynh-The et al. [39] designed a modulation classification network (MCNet) that reduces the computational complexity through the use of 2-D/3-D separable convolutions. Ma et al. [40] proposed a 3D-CNN and transformer prior network, which combines the global spatial interaction facilitated by the transformer's self-attention layer with a 3D-CNN. It effectively captures both spatial and spectral correlations of hyperspectral images, thereby addressing the limitations of CNN's learning ability. Chen et al. [41] proposed a multiscale deformable transformer, which incorporates the strengths of the CNN for local spatial–spectral information and the transformer structure for global spatial–spectral information. It can explore the diverse long-range dependencies among all spectral bands, thereby achieving higher quality reconstruction.

The research literature has revealed that deep learning-based approaches possess powerful nonlinear modeling capabilities and adaptive feature learning abilities such that they form the mainstream techniques for hyperspectral image super-resolution.

B. Limitations of Deep Learning-Based Hyperspectral Image Super-Resolution

Though deep learning-based methods have achieved success in hyperspectral image super-resolution, they generally face two challenges, which are presented as follows.

1) Indiscriminate Fusion of Features From Multiple Layers: Features from different layers of receptive fields characterize different patterns for image super-resolution. However, existing deep learning hyperspectral image super-resolution methods, especially when employing the residual network strategy, tend to indiscriminately fuse these features by summing them up via skip-connections.

Li et al. [42] proposed a grouped deep recursive residual network (GDRRN), which applies skip-connections to grouped recursive modules to capture fused feature information. Jiang et al. [43] proposed a spatial–spectral prior for super-resolution (SSPSR), which utilizes skip-connections and attention mechanisms within its spatial–spectral feature extraction block to fully exploit spatial–spectral prior information. Xu et al. [44] designed a generator based on skip-connections to enhance feature extraction capabilities. Zhang et al. [45] extracted rich local features by employing densely skip-connections convolutional layers, obtaining effective features through local feature fusion. Kong et al. [46] proposed a local feature network, which employs three convolutional layers with skip-connections local feature aggregation.

These methods normally use skip-connections to straightforwardly sum up the features generated from different

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intermediate layers. However, these features characterize different receptive fields and thus exhibit distinct hierarchical structures. The indiscriminate fusion of the features from multiple layers of a deep model gives rise to spatial information inconsistency and introduces redundant information. How to avoid the indiscriminate fusion of features and establish a more comprehensive fusion strategy is a big challenge.

2) Deficiency in Maintaining Both Spatial and Spectral Accuracy: Hyperspectral images have high spectral dimensions but limited spatial resolution. How to improve their spatial resolution in the condition of maintaining both spatial and spectral accuracy is a big challenge. Most existing methods tend to focus on one aspect while more or less underestimate the other.

Wang et al. [47] proposed a recurrent feedback super-resolution algorithm, which models spectral correlations from a sequential perspective to explore continuous spectral information. Zhang et al. [48] designed a multidimensional enhanced convolution unit, which reduces the redundancy in the spectral dimension by employing bottleneck projection, thereby condensing the useful spectral features. Hui et al. [49] proposed a spatial detail component prediction model to obtain a detail enhancement component for recovering image detail information. Xu et al. [50] designed a multiscale retinex method to clean up high-frequency noise pollution by separating the image into high-frequency and low-frequency information while enhancing the image details. Li et al. [51] proposed a dual-stage super-resolution (DualSR) method, in which the coarse-stage progressively generates super-resolved images band by band through an alternating spatial–spectral fusion mechanism.

All these methods assume that the edges in the hyperspectral images are clear and employ mapping methods to improve their spatial resolution. However, blurry edges and significant noise normally arise in the hyperspectral images. The processing procedures for super-resolution may induce further ambiguity and noise to both spatial and spectral domains. In this scenario, how to maintain both the spatial and spectral accuracy in hyperspectral image super-resolution is a big challenge.

C. Contributions of Our Work

Our novel contributions to hyperspectral image super-resolution are presented as follows.

1) In order to avoid indiscriminate feature fusion, we develop a ConvLSTM skip-connection unit. Taking advantage of the forgetting and memorization mechanisms, the ConvLSTM skip-connection unit intelligently suppresses unnecessary features and extracts useful representations for feature fusion between different layers. Therefore, it enables fusing cross-layer features in a reasonable manner.
2) In order to maintain both spatial super-resolution accuracy and spectral fidelity, we establish a framework consisting of three modules for hyperspectral image super-resolution. The three modules are all based on ConvLSTM skip-connections. They reconstruct spatial information, enhance edge details, and refine spectral information, separately. The overall framework maintains both spatial and spectral accuracy in hyperspectral image super-resolution.
3) Experimental results conducted on Chikusei, Houston, and QUST-1 datasets demonstrate that the proposed framework outperforms other methods significantly in hyperspectral image super-resolution reconstruction. We have released the code of the proposed framework to encourage public evaluation (refer to the link provided in the abstract).

II. ConvLSTM Skip-Connection

In order to demonstrate the progress from the straightforward skip-connection to our novel ConvLSTM skip-connection, the two types of skip-connections are illustrated in Fig. 1. The structure of straightforward skip-connections is illustrated in Fig. 1(a) with the one surrounded by the dashed line rectangle as a skip-connection unit. The skip-connections create short passes from one layer to another. This is an effective method for alleviating the training difficulties of deep neural networks. In skip-connections, features from different layers and receptive fields exhibit distinct patterns. Straightforward skip-connection methods may cause confusion to feature information fusion.

ConvLSTM [52] is a neural network model that integrates CNNs and long–short-term memory networks (LSTMs). Specifically, it incorporates the convolutional operations into the input, forgets, and output gates (three of the four gates except the cell gate in a normal LSTM) for characterizing spatial and sequential data.

We develop a ConvLSTM skip-connection method for comprehensively fusing feature information at different abstraction levels. The structure of sequential ConvLSTM skip-connections is illustrated in Fig. 1(b) with the one surrounded by the dashed line rectangle as a ConvLSTM skip-connection unit. Through comparing the two types of skip-connections, we observe that the skip-connection simply fuses the feature information from the preceding layer into the current layer. In contrast, the ConvLSTM skip-connection selects the feature information from the preceding layer according to the principle of long-term dependency relations through ConvLSTM. This effectively avoids the confusion in feature fusion.

The internal structure of one ConvLSTM skip-connection unit is illustrated in Fig. 2. The ConvLSTM skip-connection unit utilizes ConvLSTM to achieve skip-connections between consecutive layers. The ConvLSTM learns to selectively propagate relevant feature information to subsequent layers based on the input and past memorization. The input and forgetting gates in the ConvLSTM suppress unnecessary features, while the output gate selects useful representations to pass to deep layers. This allows for an adaptive fusion of multilevel feature information based on their abstraction and relevance. The gating mechanisms in the ConvLSTM prevent confusion in fusing features from different layers, maintaining the hierarchical nature of the learned representations.

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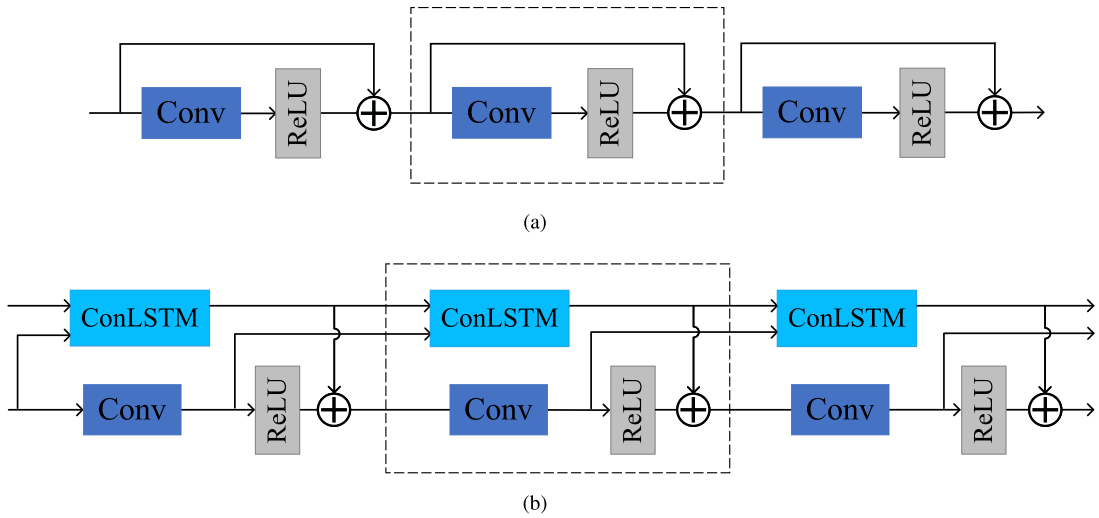


Fig. 1. Progress from (a) straightforward skip-connection to (b) our novel ConvLSTM skip-connection.

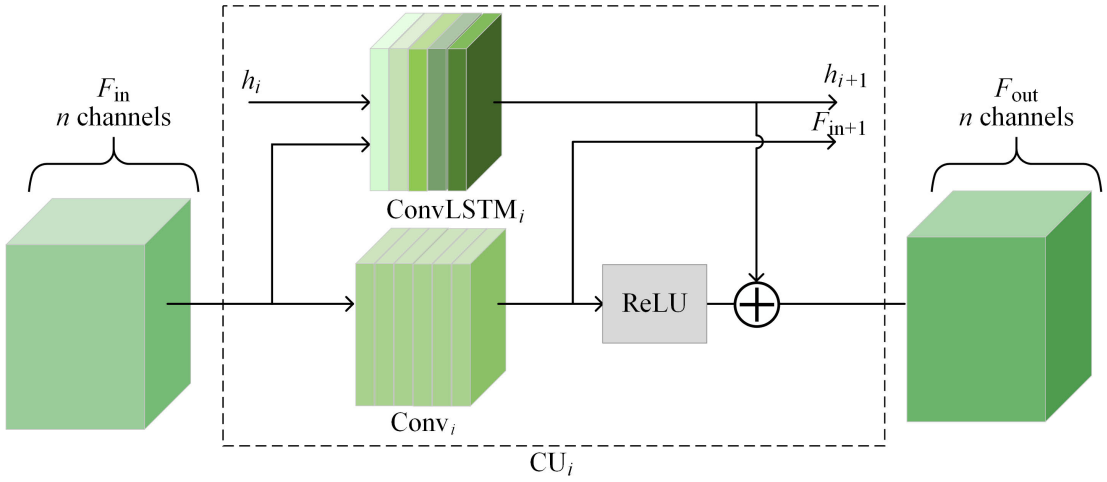


Fig. 2. ConvLSTM skip-connection. The different green cubes represent features with different levels of abstraction. CU$_i$ represents the ith ConvLSTM skip-connection unit, F_{in} represents the input feature to the CU$_i$, $F_{\text{in+1}}$ represents the input feature to the next unit, h_i represents the input memorization information to the unit, h_{i+1} represents the next memorization information from the unit, and F_{out} represents the output feature. Both input and output features have n channels.

In Fig. 2, the top part of the ConvLSTM$_i$ represents the feature selection layer, with output channels being n. ConvLSTM$_i$ capture dependency relationships and long-term memorization of features with different levels of abstraction. The bottom part of Fig. 2 consists of two components: the convolutional operator Conv$_i$ and the ReLU activation, both generating n output channels. The inputs to the ConvLSTM skip-connection unit comprise two components: the input feature F_{in} and the input memorization information h_i. The outputs F_{out} from the top and bottom parts combine through an element-wise addition operation, resulting in an output feature map with n channels. The entire process of the ConvLSTM skip-connection unit is formulated as follows:

$$F_{\text{out}} = \text{Add}(\text{ReLU}(\text{Conv}_i(F_{\text{in}})), \text{ConvLSTM}_i(h_i, F_{\text{in}})). \quad (1)$$

In the ConvLSTM skip-connection, the input feature of ConvLSTM does not pass the addition layer because it already encodes the feature representation from the previous ConvLSTM skip-connection block. Passing the input feature through addition would result in feature redundancy, weakening the current ConvLSTM memorization. In addition, the input feature of ConvLSTM in the skip-connection does not pass through activation layers. This preserves more information for the ConvLSTM gating mechanism. The gating mechanism in ConvLSTM incorporates nonlinear activation functions, allowing selective integration of input information. Applying nonlinear compression to the input features before the gating units might lead to the loss of information useful for the gating mechanism information.

III. Hyperspectral Image Super-Resolution With ConvLSTM Skip-Connections

In this section, we develop a ConvLSTM skip-connection-based hyperspectral image super-resolution framework. Our framework consists of three main modules, which complement each other to ensure excellent preservation of both spatial resolution and spectral fidelity. The first module,

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i.e., spatial feature reconstruction, employs the ConvLSTM skip-connections to comprehensively fuse spatial features from different layers. The second module, i.e., edge refinement, involves the ConvLSTM skip-connections to enhance the edge information from intermediate results. The third module, i.e., spectral information reconstruction, refines spectral features by capturing interactions between different spectral bands through the ConvLSTM skip-connections.

A. Overall Framework

As illustrated in Fig. 3, our ConvLSTM skip-connection-based hyperspectral image super-resolution framework primarily consists of three modules: the ConvLSTM skip-connection-based spatial information reconstruction, the ConvLSTM skip-connection based edge refinement, and the ConvLSTM skip-connection based spectral information reconstruction. The framework takes as input a low-resolution hyperspectral image. First, the initial feature extraction unit in the ConvLSTM skip-connection-based spatial information reconstruction module performs low-level feature extraction on the input image. The features extracted by the initial feature extraction unit are propagated to the deep feature exploration for progressively mining comprehensive features. The deep features are mapped to intermediate reconstruction results through an upsampling and reconstruction. Subsequently, the ConvLSTM skip-connection-based edge refinement module extracts edge feature information and learns elaborated edge mapping from the intermediate reconstruction results. Finally, the ConvLSTM skip-connection-based spectral information reconstruction module adjusts the spectral response using the attention values learned from the spectral guided block. This results in a final reconstructed hyperspectral image with enriched spatial–spectral detail information.

B. ConvLSTM Skip-Connection Based Spatial Information Reconstruction

Due to the high dimensionality and large volume of hyperspectral data, we adopt a progressive spatial reconstruction approach to alleviate the challenge of dimensionality. The progressive ConvLSTM skip-connection-based spatial information reconstruction consists of three stages: initial feature extraction, deep feature exploration, and upsampling and reconstruction. Its structure is illustrated in Fig. 4.

First, in the initial feature extraction stage, we employ an initial feature extraction unit, which performs the initial feature extraction of the input image through a convolutional layer

$$F_0 = \mathrm{H}_{\mathrm{ife}}(I_{\mathrm{LR}}) \tag{2}$$

where $I_{\mathrm{LR}} \in H \times W \times B$ represents the low-resolution input, and the image F_0 represents the low-level features extracted from the I_{LR} input.

The deep feature exploration stage is configured with M ConvLSTM skip-connection units. In all spatial reconstruction components, we primarily implement skip-connections within each basic unit through the feature selection layer. Let $\{\mathrm{CU}_1(\cdot), \ldots, \mathrm{CU}_{m-1}(\cdot), \mathrm{CU}_M(\cdot)\}$ represent the first ConvLSTM skip-connection unit, the $(m-1)$th ConvLSTM skip-connection unit, and the Mth ConvLSTM skip-connection unit, respectively. F_0 represents the input to the first ConvLSTM skip-connection unit. F_M represents the output to the Mth ConvLSTM skip-connection unit. The whole process of the deep feature exploration stage is expressed as

$$F_M = \mathrm{CU}_M(\mathrm{CU}_{m-1}(\cdots(\mathrm{CU}_1(F_0))\cdots)). \tag{3}$$

Finally, our framework employs a progressive approach to upsampling, gradually restoring the details of the image through multiple iterative upsampling steps. By decomposing the super-resolution reconstruction task into multiple stages, it mitigates the complexity and difficulty associated with high-dimensional data tasks. The upsampling and reconstruction are formulated as follows:

$$F_{\mathrm{SR}k} = \mathrm{Conv}_k(\mathrm{Upsampling}(F_M)) \tag{4}$$

where k represents the kth iteration of upsampling and reconstruction. Upsampling$(\cdot)$ refers to subpixel convolutional upsampling, and Conv$(\cdot)$ represents the convolutional operators. The $F_{\mathrm{SR}k}$ represents the intermediate reconstruction result of our framework. It is the intermediate reconstruction result during the kth upsampling. We add a convolutional layer immediately after the upsampling layer, which alleviates the presence of artifacts in super-resolution images.

Here, the ConvLSTM skip-connection enables spatial information reconstruction. It suppresses unnecessary spatial features and selects useful spatial representations. The selected representations are passed to deep layers, achieving feature matching during cross-layer spatial feature propagation.

C. ConvLSTM Skip-Connection Based Edge Refinement Module

Due to the excessive erosion of spatial information in hyperspectral images, directly extracting edge features from hyperspectral images gives rise to blurred edges and severe noise contamination. We develop a ConvLSTM skip-connection-based edge refinement module to extract edge features from the intermediate results $F_{\mathrm{SR}k}$ that have undergone ConvLSTM skip-connection-based spatial information reconstruction. The ConvLSTM skip-connection-based edge refinement module is illustrated in Fig. 5.

The Laplacian operator, as a typical approach, accentuates regions with the most significant grayscale variations, but it is sensitive to noise and may generate relatively coarse edges. Conversely, the Canny operator exhibits better noise resistance and often captures fine edges that may not be detected by the Laplacian operator. The edges captured by the two operators are illustrated in Fig. 6. It is observed that the two operators extract different edge information. The edges extracted by the Laplacian operator contain some noise. However, they capture edges with significant grayscale variations. The edge information extracted by the Canny operator tends to be more structured, which aids in reconstructing the structural features of the image. Therefore, our edge feature extraction combines the strengths of both the Laplacian operator and the Canny operator to capture a comprehensive range of edge details

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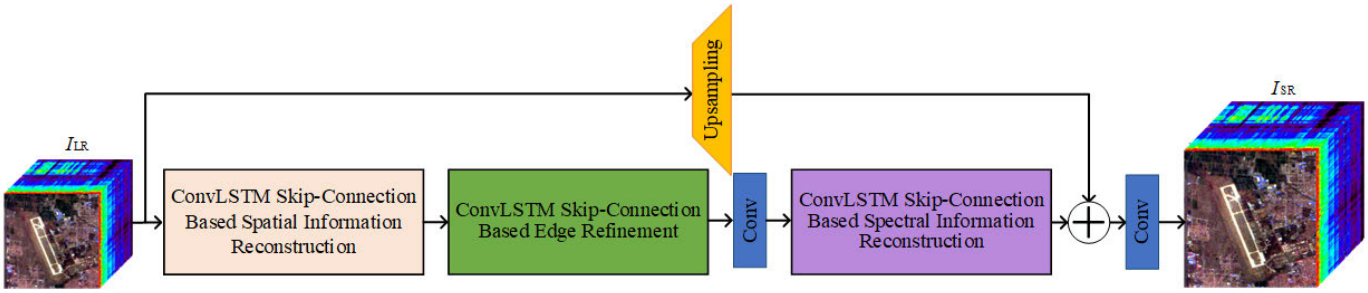


Fig. 3. Proposed ConvLSTM skip-connection based hyperspectral image super-resolution framework.

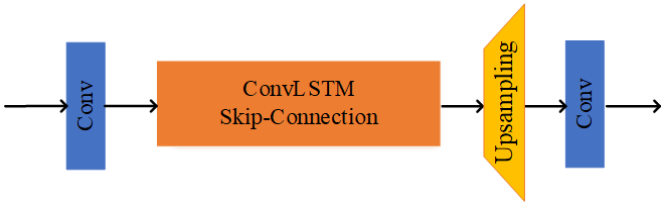


Fig. 4. ConvLSTM skip-connection based spatial information reconstruction.

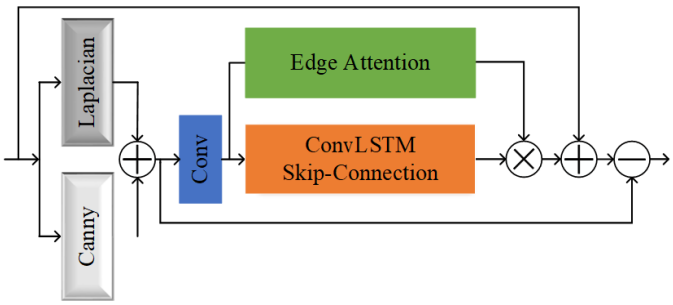


Fig. 5. ConvLSTM skip-connection based edge refinement.

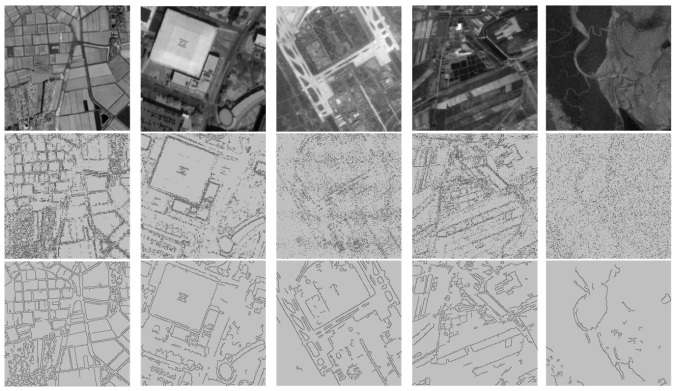


Fig. 6. Edge extraction visualization. (Top to bottom) Sequence consists of the input image, the result of edge extraction using the Laplacian operator, and the result of edge extraction using the Canny operator.

in F_{SRk}. The overall process of edge feature extraction is formulated as follows:

$$F_E = \text{Add}(\text{Laplace}(F_{SRk}), \text{Canny}(F_{SRk})) \tag{5}$$

where Laplace$(\cdot)$ represents the Laplacian operator, Canny$(\cdot)$ represents the Canny operator, and F_E represents the more comprehensive edge detail information extracted from F_{SRk}. The convolution layer delineates edge details feature. The ConvLSTM skip-connection unit is responsible for exploring deep features.

Inspired by the generative adversarial network-based edge-enhancement network [53], we develop an edge attention unit in ConvLSTM skip-connection-based edge refinement. This unit consists of multiple convolutional layers and nonlinear mapping layer pairs and one nonlinear mapping layer. The edge attention concurrently learns genuine edge weights, identifies and selects authentic edges, and filters out noisy elements, rendering fine edge reconstruction. The process of edge feature learning is given as follows:

$$F_{LE} = \text{Add}(\text{EAM}(\text{Conv}(F_E)) \otimes \text{CU}(\text{Conv}(F_E)), F_E) \tag{6}$$

where EAM$(\cdot)$ represents the edge attention mechanism operation, and F_{LE} represents learned edge features. Then, adding F_{LE} to F_{SRk} and subtracting its F_E ultimately results in a high spatial resolution space with clear and sharp edges, which is formulated as follows:

$$F_{SR'} = \text{Subtract}(\text{Add}(F_{SRk}, F_{LE}), F_E) \tag{7}$$

where Subtract$(\cdot,\cdot)$ represents the first element-by-element subtraction of the second term, and $F_{SR'}$ represents a space with high spatial resolution close to the genuine edge.

Here, the ConvLSTM skip-connection-based edge refinement module generates edge details from hyperspectral images that may suffer from blurred edges and noise pollution.

D. ConvLSTM Skip-Connection Based Spectral Information Reconstruction

Spectral dimension information is a crucial component of hyperspectral images. The ConvLSTM skip-connection-based spatial information reconstruction module and ConvLSTM skip-connection-based edge refinement module primarily focus on spatial information, paying less attention to the spectral dimension. To address this issue, we further develop the spectral information reconstruction module, which consists of P spectral guided blocks. The structure of the spectral guided block is illustrated in Fig. 7. The spectral guided block consists of two ConvLSTM skip-connection units and a spectral attention unit. The spectral information reconstruction module is used to comprehensively model the interactions between different spectral bands. The purpose of these two ConvLSTM skip-connection units is to focus more on the spectral information change during the channel transformation

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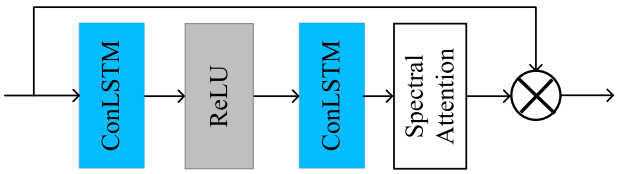


Fig. 7. ConvLSTM skip-connection based spectral information reconstruction with spectral guided block.

process. The principle of spectral attention unit is similar to channel attention. By learning attention weights during the channel expansion process, it enables adjustments to the spectral responses, ensuring the accuracy of spectral information.

Specifically, inspired by the spectral grouping and attention-driven residual dense network [36], we introduce a spectral attention unit in spectral guided block. This unit consists of multiple convolutional layers and nonlinear layers. The spectral attention unit captures spectral correlations and discriminations in the spectral data, thus favoring maintaining good spectral resolution in super-resolution tasks.

$F_q \in H \times W \times C$ is the full-band feature, where H and W indicate the spatial size and C is the number of feature channels. F_q is obtained by convolving $F_{\mathrm{SR}'}$ with a convolutional layer. By employing three linear layers, the spectral attention unit learns the coefficient relations between different spectral bands, obtaining matrix coefficient attention values, denoted as w. The gradual expansion of channels through each layer not only facilitates the capture of fundamental spectral characteristics but also contributes to the refinement of intricate feature representations. The abrupt reduction in channels allows the model to focus on selecting the most important feature channels. A spectral guided block is represented as the SGB(·), which operates in terms of wConvLSTM(ConvLSTM(·)). The calibration process of the spectral information within our framework is as follows:

$$F_{q^T} = \mathrm{SGB}_T\left(\mathrm{SGB}_{t-1}\left(\cdots\left(\mathrm{SGB}_1\left(F_q\right)\right)\cdots\right)\right) \tag{8}$$

where F_{q^T} represents the output of the Tth spectral guided block.

The element-wise addition is performed between F_{q^T} and the upsampled image of I_{LR}. The outcome of this addition is then passed through a 1×1 convolutional layer to acquire the final reconstructed image I_{SR}, according to the following:

$$I_{\mathrm{SR}} = \mathrm{Conv}\left(\mathrm{Add}\left(\mathrm{Conv}\left(F_q\right), \mathrm{Upsampling}(I_{\mathrm{LR}})\right)\right). \tag{9}$$

Here, the ConvLSTM skip-connection enables spectral information reconstruction that focuses on the spectral information change during the channel transformation process.

E. Loss Function for Training the Overall Framework

In Sections III-B–III-D, the three modules in our framework complement each other to ensure excellent preservation of both spatial resolution and spectral fidelity from hyperspectral images afflicted with blurred edge details and significant noise pollution.

Our framework is trained using the $L1$ norm as the loss function, encouraging the framework to generate high-fidelity images. The loss is given as follows:

$$\mathrm{Loss} = \frac{1}{N}\sum_{i=1}^{N}\left\|I_{\mathrm{HR}}^{i} - I_{\mathrm{SR}}^{i}\right\|_1 \tag{10}$$

where I_{SR} represents the reconstructed image obtained through learning, I_{HR} represents the original hyperspectral image, $\|\cdot\|$ represents the absolute difference between the two images calculated on a per-pixel basis, and N represents the number of training samples.

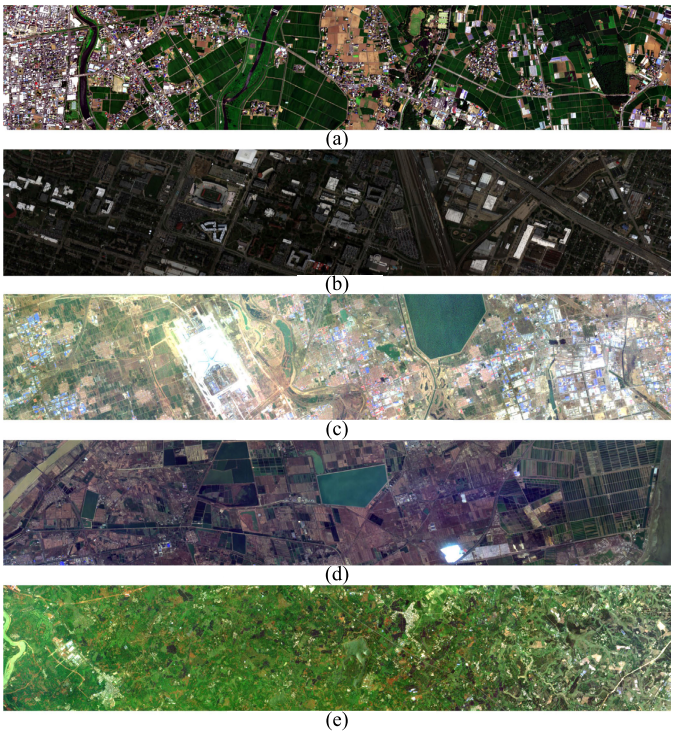


Fig. 8. Experimental data including multiple and diverse scenes. (a) Chikusei. (b) Houston. (c) QUST-1 Qingdao, China. (d) QUST-1 Dongying, Shandong. (e) QUST-1 Haikou, Hainan.

IV. Experimental Results and Discussion

A. Dataset

To evaluate the effectiveness of our proposed framework, experiments are conducted on real data from multiple different scenarios. The dataset, as illustrated in Fig. 8, consists of five hyperspectral images. These images cover a range of spatial resolutions from 1 to 10 m and include both urban and rural scenes, covering typical targets encountered in such environments.

The Chikusei hyperspectral dataset was captured on July 29, 2014. It encompasses hyperspectral image data of the agricultural and urban areas of Chikusei, Ibaraki Prefecture, Japan. It has a spatial resolution of 2.5 m and has 128 bands within the range of 363–1018 nm. The spatial dimensions of the images are 2304 × 2048 pixels.

The Houston dataset was captured on February 16, 2017. It encompasses hyperspectral image data of the Houston University campus and surrounding urban areas. The dataset has a spatial resolution of 1 m and has 48 bands within the

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range of 380–1050 nm. The spatial dimensions of the images are 4172 × 1202 pixels.

The QUST-1 satellite dataset was captured on April 10, 2019. It encompasses hyperspectral image data of Qingdao in Shandong and Haikou in Hainan, as well as Dongying in Shandong China, with one image for each location. It encompasses diverse geographical formations, including plains, mountainous regions, and oceanic areas. The spatial resolution is 10 m. It has 32 bands within the range of 400–1000 nm. Each image has a size of 5056 × 5056 pixels.

B. Evaluation Metrics

In this study, five mainstream evaluation metrics were employed to comprehensively assess the performance of different methods in terms of spatial, perceptual, and spectral aspects. These metrics include peak signal-to-noise ratio (PSNR), structural similarity (SSIM), spectral angle mapper (SAM), erreur relative globale adimensionnelle de synthèse (ERGAS), and correlation coefficient (CC). PSNR is a metric used to measure the quality of image super-resolution by comparing the pixel value differences between the generated image and the original image. However, it overlooks the sensitivity of the human eye to certain details and structures. SSIM, on the other hand, is a metric that measures image similarity. It not only considers pixel value differences but takes into account factors such as brightness, contrast, and structure. SSIM provides a better reflection of human perception of image quality. SAM is used to measure the spectral accuracy in reconstructed remote sensing images. It quantifies the angular difference of spectra at corresponding pixel locations between the reconstructed and original images to assess the consistency of spectral information. ERGAS combines spectral information with spatial resolution to evaluate the preservation of details in reconstructed images. CC reflects the geometric distortion in reconstructed images by measuring the correlation of pixel values within the image. Higher values of PSNR, SSIM, and CC indicate good performance and lower values of SAM and ERGAS indicate high quality of super-resolution.

C. Implementation Details

Each dataset is divided into training and testing data. Specifically, the scenes are divided into two nonoverlapping regions, where 80% of the images are used for training, and 20% of the images are used for testing. In addition, the dataset undergoes patch-based processing, extracting overlapping patches as reference high-resolution hyperspectral images for the training dataset. For a scale factor of 2, the extracted patches are of size 64 × 64 pixels with an overlap of 16 pixels. For a scale factor of 4, patches of the same size are used, but with an overlap of 32 pixels. In order to construct the training set, our degradation model simulates the blurring of remote sensing images by adding different disk blur kernels. The images are down-sampled using bicubic interpolation to generate low-resolution images with varying levels of blur.

We implement the proposed framework using the PyTorch framework. All experiments are conducted on a platform equipped with an NVIDIA GeForce RTX A6000 GPU and AMD EPYC 7742. In our framework, the ConvLSTM skip-connection unit M is set to 4, and the spectral guided block P is set to 3. In addition, we utilize two upsampling and reconstruction modules. During training, a learning rate decay strategy is employed to optimize the parameters of our framework. This strategy combines the Adam optimizer with the ReduceLROnPlateau learning rate scheduler. During the training process, we monitor the performance metric PSNR. When the performance does not improve in two consecutive training epochs, the learning rate is automatically reduced by a factor of 0.25, aiming to achieve better performance in subsequent training. The initial learning rate is set to 0.0002. Considering the scale of the network, we randomly select 96 low-resolution images from the training dataset as inputs for each training batch. To enhance our framework's generalization ability, we apply random rotation and flipping operations to the data.

D. Performance Evaluation

Our framework is compared with six representative hyperspectral image super-resolution methods at ×2 and ×4 upsampling factors. These methods include Bicubic, 3DFCNN [37], GDRRN [42], MCNet [39], SwinIR [28], SSPSR [43], DualSR [51], GELIN [54], and EUNet [55]. Among them, bicubic interpolation represents a traditional interpolation method. 3DFCNN and GDRRN represent classical methods based on 3-D and 2-D convolutions, respectively. MCNet stands as a state-of-the-art method based on a hybrid 2-D–3-D convolution. SwinIR is a state-of-the-art natural image super-resolution method. SSPSR is a state-of-the-art method based on residual blocks. DualSR is a state-of-the-art method employing a similar band grouping approach. GELIN is a termed group-based embedding learning and integration network. EUNet is a method that explicitly imposes the degradation model constraint. We utilize their open-source code in the PyTorch framework for comparison. For all methods, we follow the parameter settings outlined in their respective papers and conduct training using the same dataset.

1) Experimental Results on Chikusei Dataset:

a) Quantitative results: Table I summarizes the average quantitative super-resolution performance of the proposed framework and six other methods on the Chikusei dataset. The best result is highlighted in bold, and the second-best result is indicated by an underline. Our framework outperforms other methods in all evaluation metrics, including PSNR, SSIM, SAM, and ERGAS. Specifically, compared to other state-of-the-art methods, our framework exhibits improvements of 0.3028 and 0.3265 dB in PSNR at the ×2 scale and ×4 scale, respectively. In addition, the ERGAS metric decreased by 0.0112 and 0.3015 dB at the ×2 scale and ×4 scale, respectively. The SSIM and SAM metrics also exhibit improvements. This is because our framework mitigates the misalignment of high-dimensional information by standardizing spatial information fusion from different abstraction levels. In addition, our framework enhances image detail information through ConvLSTM skip-connection-based edge refinement and incorporates spectral guided block to capture spectral correlations. The bicubic performs the worst,

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TABLE I
QUANTITATIVE RESULTS OF HYPERSPECTRAL IMAGE SUPER-RESOLUTION ON THE CHIKUSEI DATASET

Scale	Metric	Bicubic	3DFCNN [37]	GDRRN [42]	MCNet [39]	SwinIR [28]	SSPSR [43]	DualSR [51]	GELIN [54]	EUNet [55]	Our framework
×2	PSNR↑	40.1653	43.3148	43.7638	44.2812	44.4703	44.5484	44.4112	44.5943	44.7235	**45.0263**
	SSIM↑	0.8622	0.9584	0.9634	0.9834	0.9812	0.9843	0.9838	0.9844	0.9856	**0.9862**
	SAM↓	3.1120	2.5333	2.2178	2.1248	2.9912	1.8690	1.9682	1.8778	1.8575	**1.8589**
	ERGAS↓	2.3167	1.8691	1.7842	1.6500	1.9378	1.4943	1.5619	1.4931	1.4878	**1.4766**
	CC↑	0.9758	0.9740	0.9740	0.9792	0.9746	0.9817	0.9822	0.9827	0.9826	**0.9846**
×4	PSNR↑	35.5357	37.9264	39.4808	39.5862	39.6987	40.2279	39.8686	40.1423	40.2534	**40.5799**
	SSIM↑	0.7235	0.9353	0.9323	0.9322	0.9236	0.9424	0.9292	0.9371	0.9429	**0.9444**
	SAM↓	3.4040	3.0779	2.6503	2.7680	3.2864	2.4575	3.1043	2.4514	2.4495	**2.4435**
	ERGAS↓	6.3495	5.8507	5.4953	4.8609	5.2374	4.0156	4.6371	4.5604	4.3364	**3.7051**
	CC↑	0.9074	0.9370	0.9474	0.9486	0.9250	0.9555	0.9448	0.9474	**0.9568**	0.9501

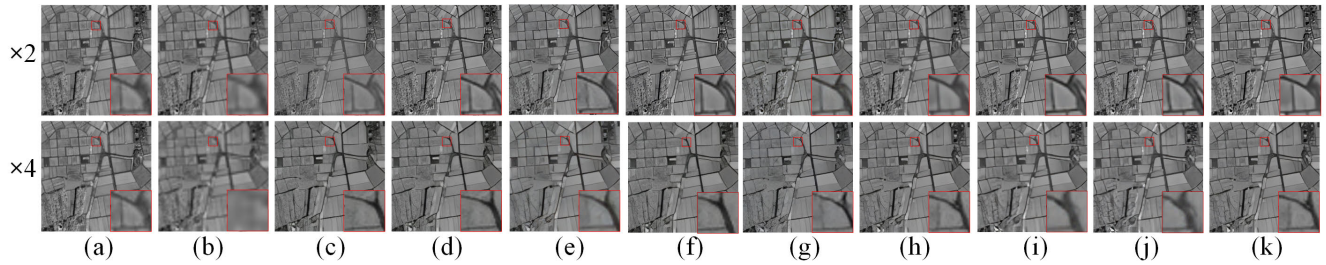


Fig. 9. Quantitative results of hyperspectral image super-resolution on the Chikusei dataset. (a) Ground truth, (b) Bicubic, (c) 3DFCNN [37], (d) GDRRN [42], (e) MCNet [39], (f) SwinIR [28], (g) SSPSR [43], (h) DualSR [51], (i) GELIN [54], (j) EUNet [55], and (k) our framework.

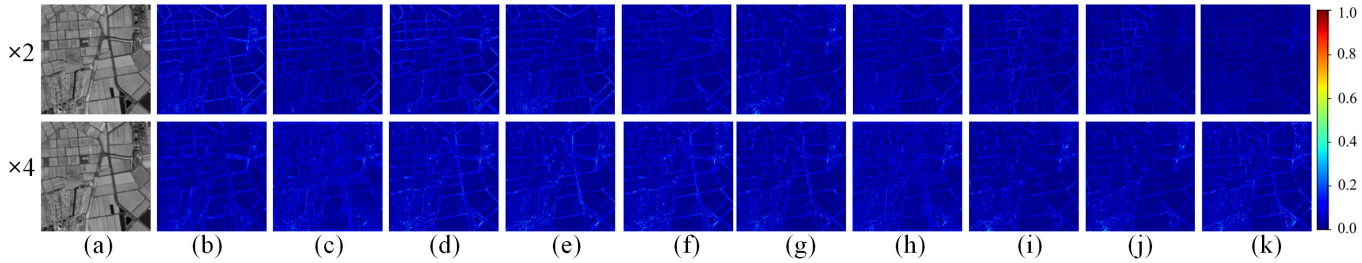


Fig. 10. Quantitative results of hyperspectral image super-resolution on the Chikusei dataset. (a) Ground truth, (b) Bicubic, (c) 3DFCNN [37], (d) GDRRN [42], (e) MCNet [39], (f) SwinIR [28], (g) SSPSR [43], (h) DualSR [51], (i) GELIN [54], (j) EUNet [55], and (k) our framework.

mainly due to its simple algorithm structure, which fails to capture sufficient spatial–spectral information. The 3DFCNN and GDRRN approaches achieve relatively better results by considering spectral correlations, but they are still limited by their model capacity. SwinIR achieves good spatial results with underestimated spectral information. Learning methods that jointly consider spatial and spectral information, such as DualSR, GELIN, and EUNet, achieve superior spatial and spectral reconstruction performance. These methods are the indiscriminative fusion of the features from multiple layers. Our framework outperforms other methods in all evaluation metrics.

b) Visual effects: Fig. 9 displays spatial super-resolution results of some test regions of the 93-band images at ×2 scale and ×4 scale, allowing for a qualitative analysis of different methods. For better visualization, the same regions marked by red rectangles are magnified. In comparison, our framework accurately restores detailed edge information, while the results of other methods appear blurry or distorted. The Bicubic method fails to generate additional details. While deep learning-based techniques such as MCNet, SwinIR, SSPSR, and DualSR reconstruct some texture details, their global optimization approaches and insufficient utilization of spatial features result in the inability to reconstruct shallower edge contours, as seen in the left target within the red box. GELIN restores details effectively, primarily because it incorporates a substantial amount of redundant information from neighboring groups into the target group to assist the reconstruction process. EUNet achieves good visual effects by alleviating the difficulty of feature learning and overfitting through disentangling spatial and spectral feature extraction. Their methods, employing straightforward skip-connection, may introduce redundant information, thus causing edge blurring. Specifically, our framework produces the most visually realistic and clear results.

Fig. 10 displays the absolute error maps of the comparative methods for the 93-band images at ×2 scale and ×4 scale. These maps reflect the differences between the reconstructed images and the ground truth images, where a small error mapping indicates the high performance of the SR methods.

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TABLE II
QUANTITATIVE RESULTS OF HYPERSPECTRAL IMAGE SUPER-RESOLUTION ON THE HOUSTON DATASET

Scale	Metric	Bicubic	3DFCNN [37]	GDRRN [42]	MCNet [39]	SwinIR [28]	SSPSR [43]	DualSR [51]	GELIN [54]	EUNet [55]	Our framework
×2	PSNR↑	36.4386	38.6159	38.3290	39.1909	39.1744	39.4546	39.5536	39.5449	39.5776	**40.1882**
	SSIM↑	0.9246	0.9754	0.9616	0.9772	0.9575	0.9800	0.9793	0.9811	0.9813	**0.9845**
	SAM↓	2.2523	2.6916	2.1501	1.9003	3.0957	1.6264	1.6699	1.6528	1.6114	**1.4571**
	ERGAS↓	1.3575	1.0354	0.7460	0.6701	1.2630	0.5936	0.6077	0.5663	0.5716	**0.5488**
	CC↑	0.9668	0.9790	0.9888	0.9905	0.9846	0.9922	0.9918	0.9920	0.9923	**0.9939**
×4	PSNR↑	32.2735	35.1125	35.5620	35.9246	35.6718	36.2978	36.2602	36.2364	36.3111	**36.5753**
	SSIM↑	0.7262	0.9192	0.9243	0.9337	0.8862	0.9376	0.9368	0.9355	0.9380	**0.9385**
	SAM↓	2.6523	2.4843	2.3023	2.2078	2.5801	2.1812	2.0224	2.0247	2.0209	**2.0190**
	ERGAS↓	2.8862	1.1859	1.1274	1.0829	1.8240	1.0331	1.0144	1.0207	1.0180	**1.0003**
	CC↑	0.9561	0.9686	0.9714	0.9750	0.9692	0.9766	0.9764	0.9763	0.9768	**0.9778**

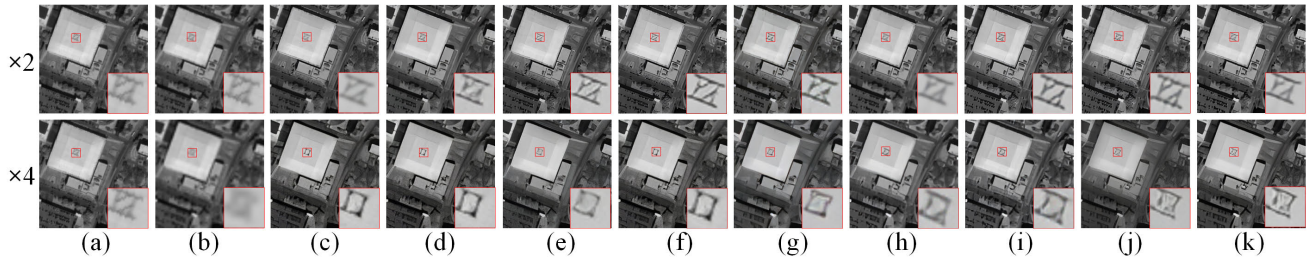


Fig. 11. Quantitative results of hyperspectral image super-resolution on the Houston dataset. (a) Ground truth, (b) Bicubic, (c) 3DFCNN [37], (d) GDRRN [42], (e) MCNet [39], (f) SwinIR [28], (g) SSPSR [43], (h) DualSR [51], (i) GELIN [54], (j) EUNet [55], and (k) our framework.

In Fig. 10, the images reconstructed by our framework exhibit small errors. Our framework achieves the best spatial fidelity in terms of reconstructing the spatial information in the images. Specifically, our framework demonstrates outstanding performance in reconstructing the edges of field pathways and the textures of the fields. This is likely attributed to the ConvLSTM skip-connection-based edge refinement module in our framework, which reduces the influence of edge noise on the imaging process, resulting in high spatial fidelity in the reconstructed images.

We visualize the spectral curves of selected pixels (indicated by red dots in the image) on the reconstructed images from the Chikusei dataset, with an image at ×4 scale, as shown in Fig. 19(a). It is evident that in most cases, the spectral curves of our framework are closer to the ground truth values compared to those of the comparative methods. Therefore, the advantage of our framework in spectral preservation is clearly illustrated, confirming the effectiveness of the spectral information reconstruction module.

The relatively simple texture of the Chikusei dataset is due to the fact that most of the land is covered by various vegetation, with more fields and rivers, and fewer buildings. Our framework effectively restores edge and structural information, resulting in minimal differences between the generated field images and the original images. This indicates that our proposed framework exhibits strong super-resolution performance when dealing with regular natural pattern data.

2) Experimental Results on Houston Dataset:

a) Quantitative results: Table II summarizes the average quantitative super-resolution performance of our framework and six other methods on the Houston dataset. The best results are highlighted in bold, while the second-best results are indicated with an underline. It is observed that our framework achieves good performance at both ×2 scale and ×4 scale, with PSNR values of 40.1882 and 36.5753 dB, SSIM values of 0.9845 and 0.9385, SAM values of 1.4571 and 2.0190, ERGAS values of 0.5488 and 1.0003, and CC values of 0.9939 and 0.9778, respectively. Compared to state-of-the-art methods, our approach achieves an improvement of 0.6106 and 0.2642 dB in terms of PSNR, and an enhancement of 0.0448 and 0.0141 in ERGAS, respectively.

b) Visual effects: Fig. 11 presents the visualizations of different super-resolution methods at ×2 and ×4 scales, displayed in the 39-band representation. To facilitate evaluation, key areas are magnified and highlighted with red rectangular boxes in the figure. The red rectangular regions reveal that the symbol edge information in the super-resolutions by Bicubic, 3DFCNN, and GDRRN appears blurry. The edges reconstructed by MCNet and DualSR exhibit artifacts, while the results of SSPSR appear overly sharpened. Compared with GELIN and EUNet, our results exhibit clear edges and contour structures that closely resemble the ground truth. Our framework is capable of recovering more edge and contour information, resulting in superior spatial super-resolution results.

Fig. 12 displays the absolute error maps of the comparative methods for the 39-band images at ×2 and ×4 scales. In Fig. 12, the images reconstructed by our framework exhibit the smallest errors. Our framework demonstrates outstanding performance in reconstructing high-frequency edges and low-frequency textures. Fig. 19(b) illustrates the spectral curves of randomly selected pixels (indicated by red dots in the image) on the reconstructed images from the Houston dataset, with an image scale factor of 4. It is evident that our framework

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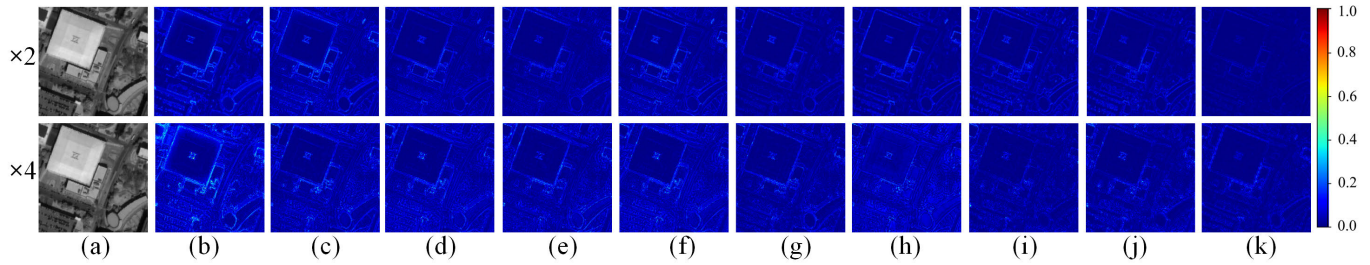


Fig. 12. Quantitative results of hyperspectral image super-resolution on the Houston dataset. (a) Ground truth, (b) Bicubic, (c) 3DFCNN [37], (d) GDRRN [42], (e) MCNet [39], (f) SwinIR [28], (g) SSPSR [43], (h) DualSR [51], (i) GELIN [54], (j) EUNet [55], (k) Our framework.

TABLE III
QUANTITATIVE RESULTS OF HYPERSPECTRAL IMAGE SUPER-RESOLUTION ON THE QUST-1 DATASET

Scale	Metric	Bicubic	3DFCNN [37]	GDRRN [42]	MCNet [39]	SwinIR [28]	SSPSR [43]	DualSR [51]	GELIN [54]	EUNet [55]	Our framework
×2	PSNR↑	34.5754	36.1629	37.7633	39.5551	38.4578	40.5205	40.6056	40.7265	40.8354	**41.3523**
	SSIM↑	0.9689	0.9739	0.9799	0.9837	0.9639	0.9849	0.9859	0.9865	0.9874	**0.9894**
	SAM↓	2.0908	1.7682	1.5503	1.3291	1.9341	1.2476	1.2001	1.1996	1.1893	**1.1727**
	ERGAS↓	0.6140	0.5226	0.4341	0.3536	0.5403	0.3299	0.3169	0.3154	0.3149	**0.3103**
	CC↑	0.8338	0.8809	0.9091	0.9369	0.9385	0.9440	0.9482	0.9481	0.9498	**0.9510**
×4	PSNR↑	30.1947	34.9671	34.4411	35.8468	34.9470	35.9957	35.9208	36.1200	36.3680	**36.6229**
	SSIM↑	0.8234	0.9424	0.9524	0.9551	0.9481	0.9649	0.9711	0.9723	0.9717	**0.9752**
	SAM↓	2.6553	2.0876	1.7856	1.7125	2.2653	1.4417	1.5338	1.4781	1.4287	**1.3977**
	ERGAS↓	1.7679	1.3369	0.9028	0.8457	1.6231	0.4299	0.8520	0.4244	0.4232	**0.4185**
	CC↑	0.8758	0.8925	0.9035	0.9130	0.8975	0.9127	0.9116	0.9139	0.9133	**0.9150**

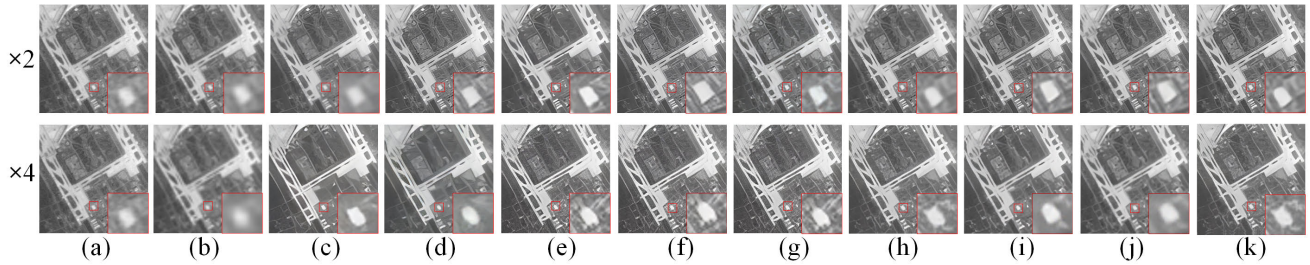


Fig. 13. Quantitative results of hyperspectral image super-resolution on the QUST-1 Qingdao dataset. (a) Ground truth, (b) Bicubic, (c) 3DFCNN [37], (d) GDRRN [42], (e) MCNet [39], (f) SwinIR [28], (g) SSPSR [43], (h) DualSR [51], (i) GELIN [54], (j) EUNet [55], and (k) our framework.

maintains superior results compared to other methods while achieving high spectral fidelity.

The Houston dataset has a higher spatial resolution compared to the other two datasets, featuring larger ground objects, rich textures, and relatively regular buildings. Our proposed framework utilizes this abundant information to reconstruct fine details in the super-resolved images, thereby reducing the gap between the reconstructed and original images. It is observed that our proposed framework is applicable for handling artificial pattern data with rich texture details.

3) Experimental Results on QUST-1 Dataset:

a) Quantitative results: Table III summarizes the average quantitative super-resolution performance of six methods conducted on three image datasets from QUST-1. The best results are highlighted in bold, while the second-best results are indicated with an underline. From Table III, it is observed that 3DFCNN and GDRRN, due to their shallow model architectures, have limited expressive power, resulting in poor performance. Although MCNet increases depth through recursive structures, the reduction in parameter quantity through dimensionality reduction steps limits the model's capacity. In contrast, the high complexity models, SSPSR and DualSR, demonstrate significant advantages, with substantial improvements in quantitative results. This underscores the importance of model architecture for hyperspectral image super-resolution. GELIN and EUNet demonstrate, through excellent model design, that low-complexity models can also produce high-quality results. Our framework achieves the best values in all metrics for both scales, outperforming the other methods. This is because our framework facilitates the fusion of features at different abstraction levels and learns real and effective edge information.

b) Visual effects: Figs. 13–15 depict the visualizations of different super-resolution methods on the Qingdao, Dongying, and Kaikou images from the QUST-1 dataset at ×2 and ×4 scales, displayed in 3 bands. The red rectangular regions in the images contain abundant edge and texture information, posing a challenge for hyperspectral image super-resolution reconstruction. In Fig. 13, the objects are relatively large, and the edge structure information is abundant. Specifically, focusing on the red rectangular region in Fig. 13, the GDRRN,

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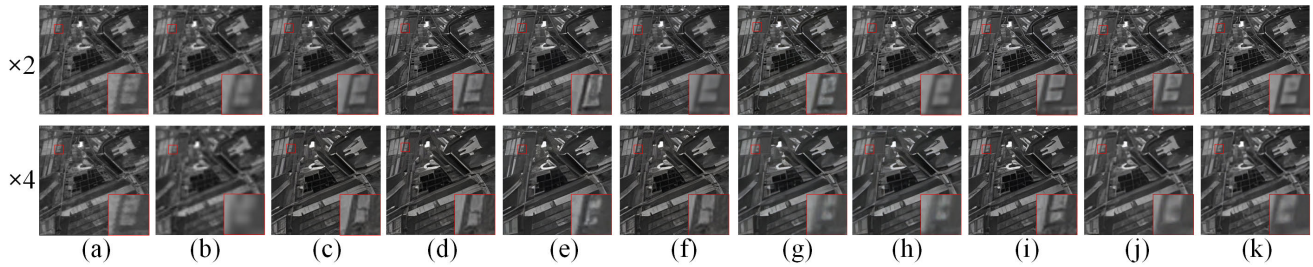


Fig. 14. Quantitative results of hyperspectral image super-resolution on the QUST-1 Dongying dataset. (a) Ground truth, (b) Bicubic, (c) 3DFCNN [37], (d) GDRRN [42], (e) MCNet [39], (f) SwinIR [28], (g) SSPSR [43], (h) DualSR [51], (i) GELIN [54], (j) EUNet [55], and (k) our framework.

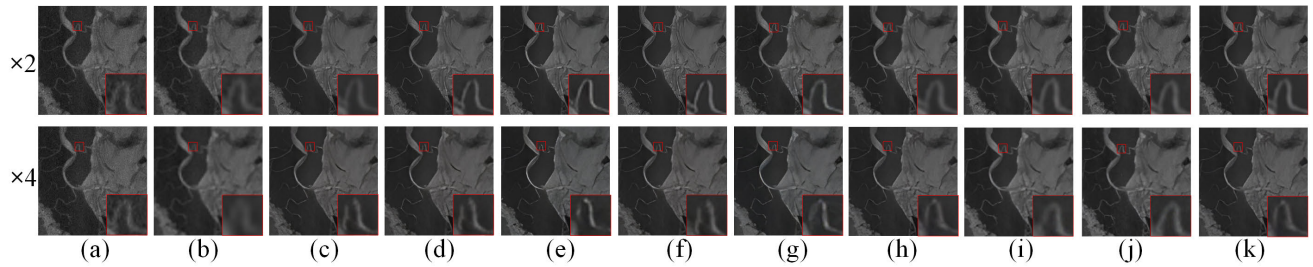


Fig. 15. Quantitative results of hyperspectral image super-resolution on the QUST-1 Haikou dataset. (a) Ground truth, (b) Bicubic, (c) 3DFCNN [37], (d) GDRRN [42], (e) MCNet [39], (f) SwinIR [28], (g) SSPSR [43], (h) DualSR [51], (i) GELIN [54], (j) EUNet [55], and (k) our framework.

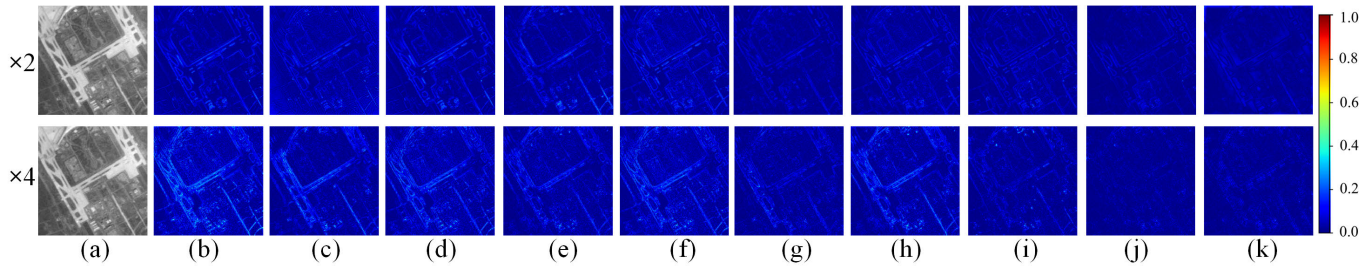


Fig. 16. Absolute error maps for the QUST-1 Qingdao dataset. (a) Ground truth, (b) Bicubic, (c) 3DFCNN [37], (d) GDRRN [42], (e) MCNet [39], (f) SwinIR [28], (g) SSPSR [43], (h) DualSR [51], (i) GELIN [54], (j) EUNet [55], and (k) our framework.

MCNet, and DualSR methods exhibit excessively smoothed contours in their super-resolution methods, while the results from 3DFCNN and SSPSR have image distortion. In contrast, GELIN and EUNet achieve heavier super-resolution visual artifacts. Our framework enhances the clarity of the image and endows it with strong adaptability. In Fig. 14, the objects are relatively small, and there is abundant texture information. For the red rectangular region in Fig. 14, our framework is still capable of reconstructing a clear and artifact-free image. In Fig. 15, the content mainly consists of irregular natural river edges. The super-resolution methods by 3DFCNN, GDRRN, and DualSR exhibit blurry edge information, while MCNet and SSPSR have artifacts and distortions in their super-resolution methods. In contrast, our framework's super-resolution results are closest to the original image, demonstrating that our framework is more suitable for reconstructing irregular natural images. This observation is further supported by the absolute error maps in Figs. 16–18. Fig. 19(c) presents the spectral curves of randomly selected pixels (indicated by red dots in the image) on the reconstructed images from QUST-1, with an image scale factor of 4. It is evident from the figure that our framework continues to exhibit robust performance.

The QUST-1 dataset is characterized by a large volume of data and diverse scenes with various land cover types and geographical forms. Our proposed framework effectively reconstructs both regular artificial patterns and irregular natural forms present in these scenes. It is validated that our framework is suitable for handling diverse scene data and exhibits strong robustness.

E. Ablation Studies

In this section, to validate the effectiveness of our proposed framework, we conduct extensive ablation experiments on the QUST-1 dataset with an image upsampling factor of 4. These experiments focus on four aspects, including the manner of skip-connections, the number of ConvLSTM skip-connection units for deep feature exploration, the placement of the ConvLSTM skip-connection-based edge refinement module, and the edge extraction method employed by the ConvLSTM skip-connection-based edge refinement module.

1) Manner of Skip-Connections: Table IV reports the impact of different skip-connection methods on the QUST-1 dataset. NCLSCU represents our framework without a ConvLSTM unit, where the features between two layers are directly skip-connected. DSCU represents using a convolutional layer as a replacement for ConvLSTM in the skip-connection. CLSCU represents our framework. From Table IV, it is observed that our framework achieves the best performance,

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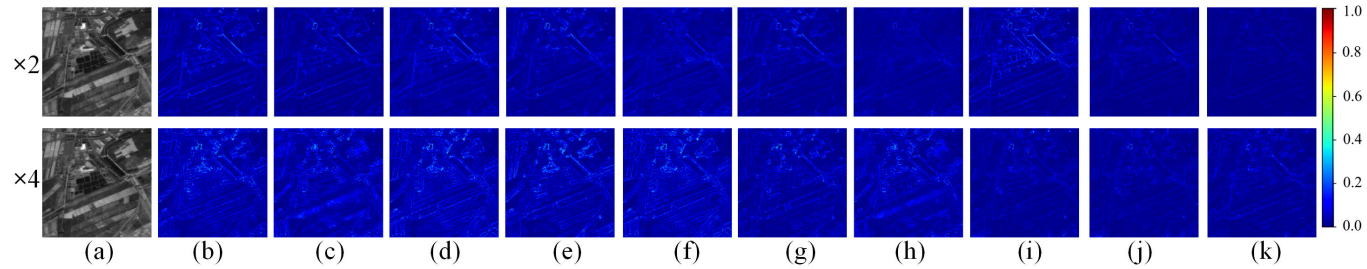


Fig. 17. Absolute error maps for the QUST-1 Dongying dataset. (a) Ground truth, (b) Bicubic, (c) 3DFCNN [37], (d) GDRRN [42], (e) MCNet [39], (f) SwinIR [28], (g) SSPSR [43], (h) DualSR [51], (i) GELIN [54], (j) EUNet [55], and (k) our framework.

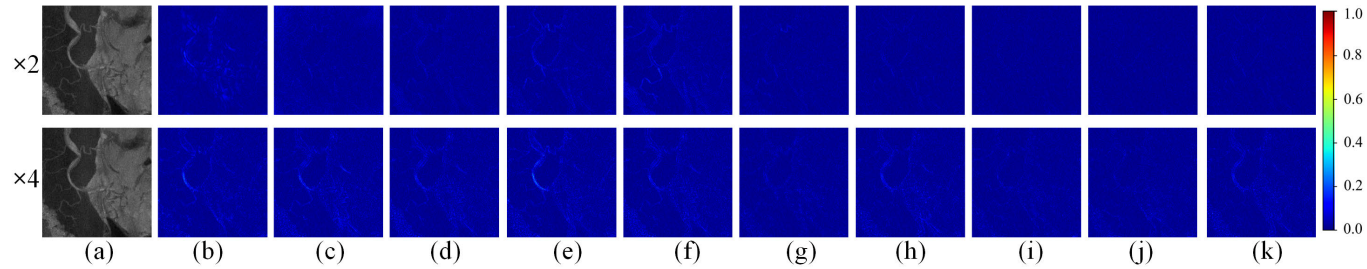


Fig. 18. Absolute error maps for the QUST-1 Haikou dataset. (a) Ground truth, (b) Bicubic, (c) 3DFCNN [37], (d) GDRRN [42], (e) MCNet [39], (f) SwinIR [28], (g) SSPSR [43], (h) DualSR [51], (i) GELIN [54], (j) EUNet [55], and (k) our framework.

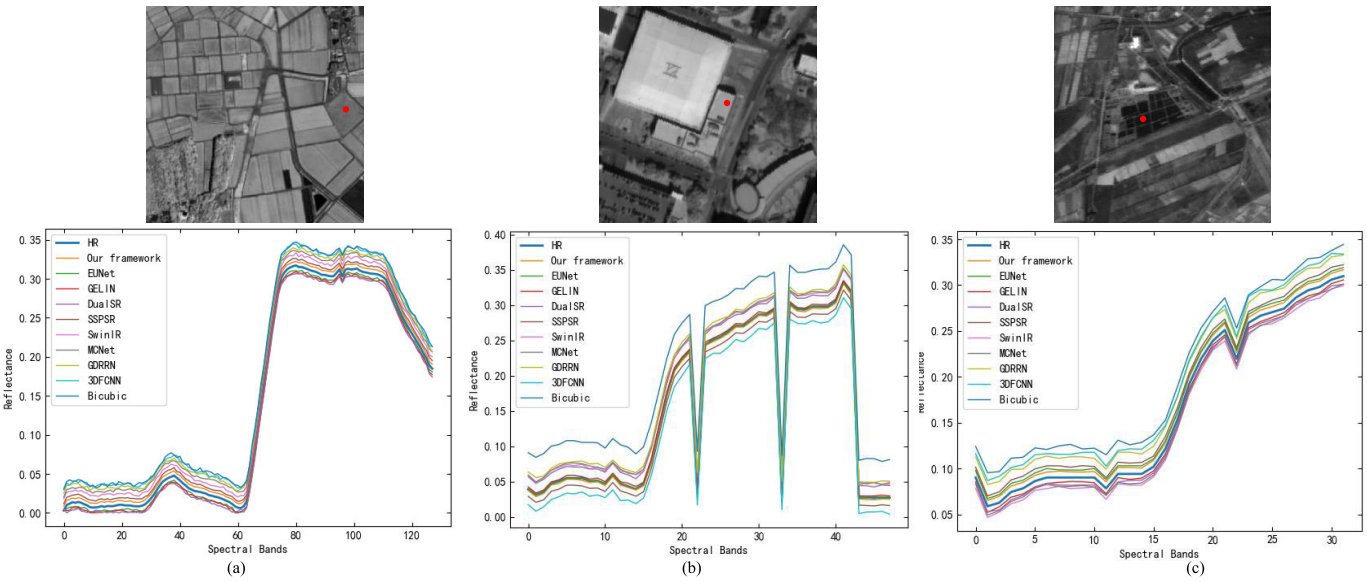


Fig. 19. Spectral curves of selected locations on (a) Chikusei, (b) Houston, and (c) QUST-1 datasets.

TABLE IV
RESULTS OF DIFFERENT SKIP-CONNECTION MANNERS ON THE QUST-1 DATASET

Type	PSNR↑	SSIM↑	SAM↓	ERGAS↓	CC↑
NCLSCU	35.5409	0.9315	1.6422	0.5912	0.8715
DSCU	35.9638	0.9677	1.4526	0.4499	0.9127
CLSCU	**36.6229**	**0.9752**	**1.3977**	**0.4185**	**0.9150**

while the model using NCLSCU performs the worst in terms of the metrics. This demonstrates the necessity of standardizing features of different abstraction levels and highlights the superior performance of the ConvLSTM skip-connection unit in this regard.

2) Number of ConvLSTM Skip-Connection Units for Deep Feature Exploration: Table V reports the impact of different numbers of ConvLSTM skip-connection units for deep feature exploration on the QUST-1 dataset. Five evaluation metrics are employed for assessment. From Table V, it is observed that the results are the worst when there are three ConvLSTM skip-connection units. This is mainly due to the insufficient depth of the framework, which fails to adequately capture spatial information. When there are four ConvLSTM skip-connection units, the spatial metrics perform the best. When the number of ConvLSTM skip-connection units is five or six, the reconstruction results of the model are essentially the same. Therefore, considering the tradeoff between reconstruction

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TABLE V
RESULTS OF DIFFERENT NUMBERS OF CONVLSTM SKIP-CONNECTION UNITS FOR DEEP FEATURE EXPLORATION ON THE QUST-1 DATASET

Different number of ConvLSTM skip-connection units	3	4	5	6
PSNR↑	34.9149	**36.6229**	36.4216	36.4205
SSIM↑	0.8752	**0.9752**	0.9652	0.9580
SAM↓	2.2357	1.3977	**1.4126**	1.4213
ERGAS↓	0.6314	0.4185	**0.4129**	0.4169
CC↑	0.8245	**0.9150**	0.9141	0.9141

TABLE VI
RESULTS OF DIFFERENT PLACEMENTS OF EDGE REFINEMENT ON MODULES THE QUST-1 DATASET

Type	PSNR↑	SSIM↑	SAM↓	ERGAS↓	CC↑
ERN-Front	35.897	0.9488	1.4641	0.4266	0.9015
ERN-Middle	**36.6229**	**0.9752**	**1.3977**	**0.4185**	**0.9150**
ERN-After	36.007	0.9530	1.5125	0.4387	0.8899

TABLE VII
RESULTS OF DIFFERENT EDGE FEATURE EXTRACTION METHODS ON THE QUST-1 DATASET

Type	PSNR↑	SSIM↑	SAM↓	ERGAS↓	CC↑
ERN-Lap	35.9492	0.9749	1.4322	0.4231	0.9131
ERN-Canny	35.9330	0.9747	1.4329	0.4235	0.9130
ERN	**36.6229**	**0.9752**	**1.3977**	**0.4185**	**0.9150**

performance and training cost, the number of ConvLSTM skip-connection units is set to four in the experiments.

3) Placement of the ConvLSTM Skip-Connection Based Edge Refinement Module: Table VI reports the impact of different placements of the ConvLSTM skip-connection-based edge refinement module on the QUST-1 dataset. ERN-Front indicates that the ConvLSTM skip-connection-based edge refinement module is located before the initial feature extraction unit, ERN-Middle indicates that the ConvLSTM skip-connection-based edge refinement module is placed after the last reconstruction module, and ERN-After indicates that the ConvLSTM skip-connection based edge refinement module is located at the end of the entire model. From Table VI, it can be observed that ERN-Middle achieves the highest reconstruction quality, while ERN-Front has the lowest PSNR value. This could be attributed to the limited presence of edge texture information in the low-resolution hyperspectral image, which restrains the ConvLSTM skip-connection-based edge refinement module from learning sharp edge information effectively. On the other hand, ERN-After exhibits the worst SAM performance, which is possibly due to ERN-After disrupting the spectral responses that have already been adjusted by the spectral attention unit.

4) Edge Feature Extraction Method of the ConvLSTM Skip-Connection Based Edge Refinement Module: From Table VII we can observe the impact of different edge extraction methods employed by the ConvLSTM skip-connection-based edge refinement module on the QUST-1 dataset. ERN-Lap indicates the use of the Laplacian operator for edge extraction, ERN-Canny represents the adoption of the Canny operator to extract edge information, and ERN indicates the joint utilization of both the Laplacian operator and Canny operator to extract edge information. The extracted edge information is fused by weighted fusion with equal weights of 0.5. From Table VII, it is observed that the model utilizing the ERN module achieves the best evaluation quality. This reflects that ERN is capable of providing abundant edge information to the model, which is effectively utilized, thereby enabling the model to achieve superior learning performance.

V. CONCLUSION

This article has introduced a ConvLSTM skip-connection strategy to characterize features from consecutive layers using ConvLSTMs. Compared with straightforward skip-connection methods, the ConvLSTM skip-connection strategy comprehensively fuses feature information at different abstraction levels. This strategy avoids the inconsistency of spatial information and the introduction of redundant information.

Based on the ConvLSTM strategy, we have developed a novel hyperspectral image super-resolution framework. The framework comprises three modules: the ConvLSTM skip-connection-based spatial information reconstruction, the ConvLSTM skip-connection-based edge refinement, and the ConvLSTM skip-connection-based spectral information reconstruction. First, the spatial information is reconstructed using the method based on ConvLSTM skip-connection. This approach facilitates the incorporation of hierarchical information, thereby achieving remarkable spatial restoration performance. Second, the ConvLSTM skip-connection-based edge refinement module extracts spatial edge information from the generated images. This spatial edge information is learned and filtered to reconstruct detailed edge information. Third, we have introduced a spectral information reconstruction component to comprehensively model interactions between different spectral bands. By adjusting spectral responses within spectral guided blocks through learned attention weights, the spectral fidelity is maintained. The combination of its three modules has achieved the utilization of features from different levels to reconstruct hyperspectral images that closely resemble ground truth. Our framework has provided visually superior super-resolution images. A comprehensive evaluation, including qualitative and quantitative assessments, has validated that our framework outperforms six other hyperspectral image super-resolution methods.

ConvLSTM skip-connection is a general structure for deep learning. It will not be confined to the use of hyperspectral image super-resolution but can be explored for broader applications such as denoising, unmixing, and classification.

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